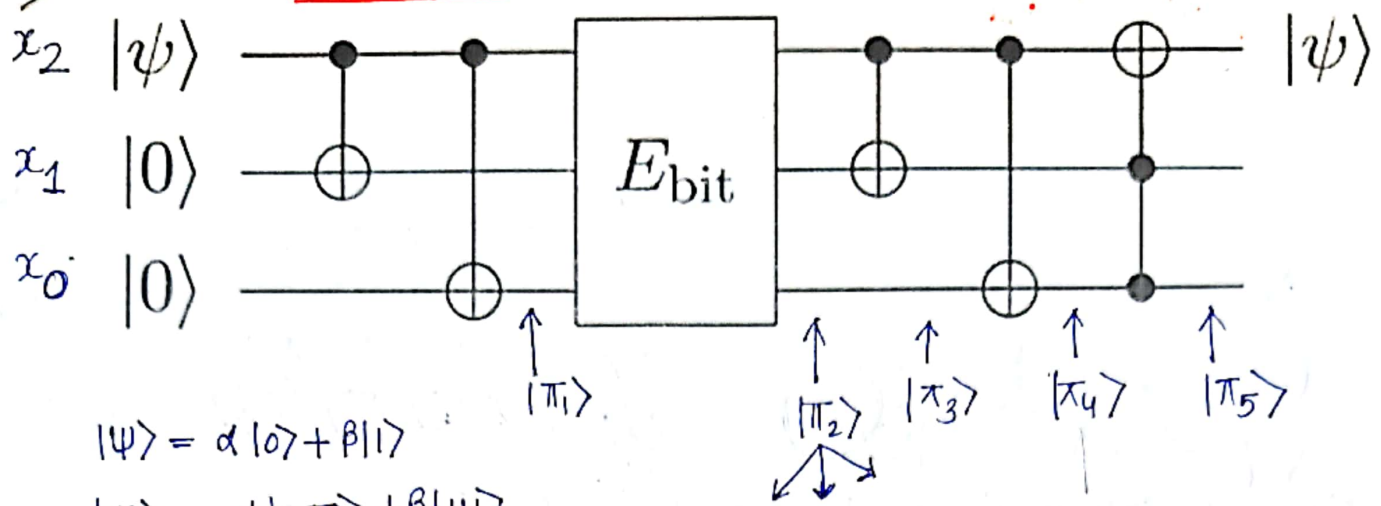


Bit-flip



$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

$$|\pi_1\rangle = \alpha|000\rangle + \beta|111\rangle$$

$$|\pi_2\rangle = \alpha|100\rangle + \beta|011\rangle \text{ OR } \alpha|010\rangle + \beta|101\rangle \text{ OR } \alpha|001\rangle + \beta|110\rangle$$

$$|\pi_2\rangle = \alpha|100\rangle + \beta|011\rangle$$

$$|\pi_3\rangle = \text{CNOT}_{21}(\alpha|100\rangle + \beta|011\rangle) = \alpha|110\rangle + \beta|011\rangle$$

$$|\pi_4\rangle = \text{CNOT}_{20}(\alpha|110\rangle + \beta|011\rangle) = \alpha|111\rangle + \beta|011\rangle$$

$$|\pi_5\rangle = \text{CCNOT}_{012}(\alpha|111\rangle + \beta|011\rangle) = (\alpha|011\rangle + \beta|111\rangle) \otimes |11\rangle$$

$$|\pi_2\rangle = \alpha|010\rangle + \beta|101\rangle$$

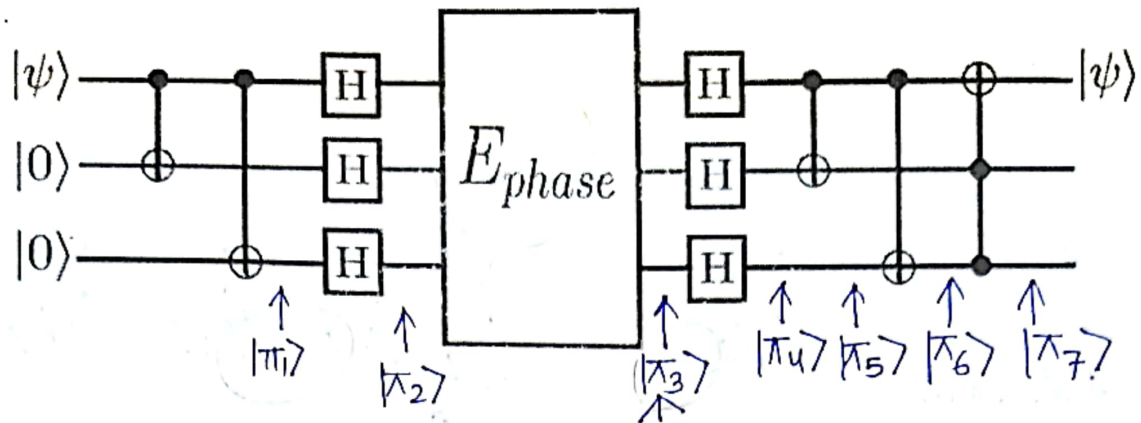
$$|\pi_3\rangle = \text{CNOT}_{21}(\alpha|010\rangle + \beta|101\rangle) = \alpha|010\rangle + \beta|111\rangle$$

$$|\pi_4\rangle = \text{CNOT}_{20}(\alpha|010\rangle + \beta|111\rangle) = \alpha|010\rangle + \beta|110\rangle$$

$$|\pi_5\rangle = \text{CCNOT}_{012}(\alpha|010\rangle + \beta|110\rangle) = \alpha|010\rangle + \beta|110\rangle$$

$$= (\alpha|0\rangle + \beta|1\rangle) \otimes |10\rangle$$

Phase-flip



$$|\pi_1\rangle = \alpha |000\rangle + \beta |111\rangle \quad \& \quad |\pi_2\rangle = \alpha |+++ \rangle + \beta |-- \rangle$$

Lets assume, $|\pi_3\rangle = \alpha |-++ \rangle + \beta |+-- \rangle$

then $|\pi_4\rangle = \alpha |100\rangle + \beta |011\rangle \quad \& \quad |\pi_5\rangle = \alpha |110\rangle + \beta |011\rangle$

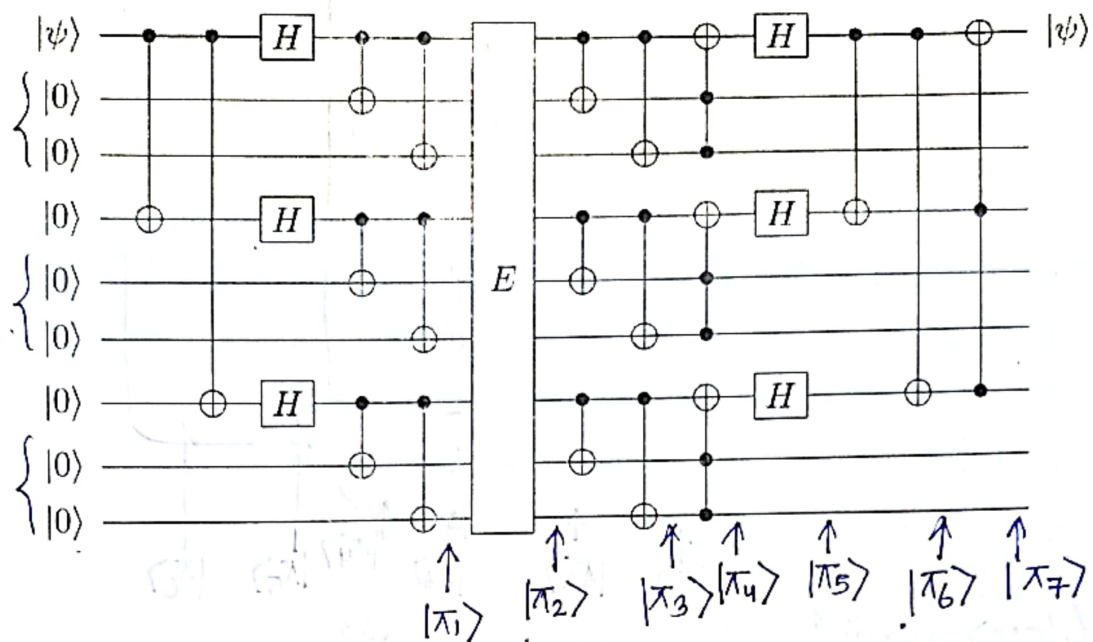
$$|\pi_6\rangle = \alpha |111\rangle + \beta |011\rangle$$

$$|\pi_7\rangle = \alpha |011\rangle + \beta |111\rangle$$

$$= (\alpha |0\rangle + \beta |1\rangle) \otimes |11\rangle$$

Shor's 9 qubit

Single
bit-flip
error



$$|\pi_1\rangle = \frac{1}{\sqrt{2}} \alpha (|1000\rangle + |1111\rangle) \otimes (|1000\rangle + |1111\rangle) \otimes (|1000\rangle + |1111\rangle) \\ + \frac{1}{\sqrt{2}} \beta (|1000\rangle - |1111\rangle) \otimes (|1000\rangle - |1111\rangle) \otimes (|1000\rangle - |1111\rangle)$$

$$|\pi_2\rangle = \frac{1}{\sqrt{2}} \alpha (|1100\rangle + |0111\rangle) \otimes (|1000\rangle + |1111\rangle) \otimes (|1000\rangle + |1111\rangle) \\ + \frac{1}{\sqrt{2}} \beta (|1100\rangle - |0111\rangle) \otimes (|1000\rangle - |1111\rangle) \otimes (|1000\rangle - |1111\rangle)$$

$$|\pi_3\rangle = \frac{1}{\sqrt{2}} \alpha (|1111\rangle + |0111\rangle) \otimes (|1000\rangle + |1000\rangle) \otimes (|1000\rangle + |1000\rangle) \\ + \frac{1}{\sqrt{2}} \beta (|1111\rangle - |0111\rangle) \otimes (|1000\rangle - |1000\rangle) \otimes (|1000\rangle - |1000\rangle)$$

$$|\pi_4\rangle = \frac{1}{\sqrt{2}} \alpha (|0111\rangle + |1111\rangle) \otimes (|1000\rangle + |1000\rangle) \otimes (|1000\rangle + |1000\rangle) \\ + \frac{1}{\sqrt{2}} \beta (|0111\rangle - |1111\rangle) \otimes (|1000\rangle - |1000\rangle) \otimes (|1000\rangle - |1000\rangle)$$

$$|\pi_5\rangle = \frac{1}{\sqrt{2}} \alpha (|107\rangle \otimes |1117\rangle) \otimes (|107\rangle \otimes |1007\rangle) \otimes (|107\rangle \otimes |1007\rangle) \\ + \frac{1}{\sqrt{2}} \beta (|107\rangle \otimes |1117\rangle) \otimes (|107\rangle \otimes |1007\rangle) \otimes (|107\rangle \otimes |1007\rangle)$$

$$|\pi_6\rangle = \frac{1}{\sqrt{2}} \alpha (|107\rangle \otimes |1117\rangle) \otimes (|107\rangle \otimes |1007\rangle) \otimes (|107\rangle \otimes |1007\rangle) \\ + \frac{1}{\sqrt{2}} \beta (|107\rangle \otimes |1117\rangle) \otimes (|107\rangle \otimes |1007\rangle) \otimes (|107\rangle \otimes |1007\rangle)$$

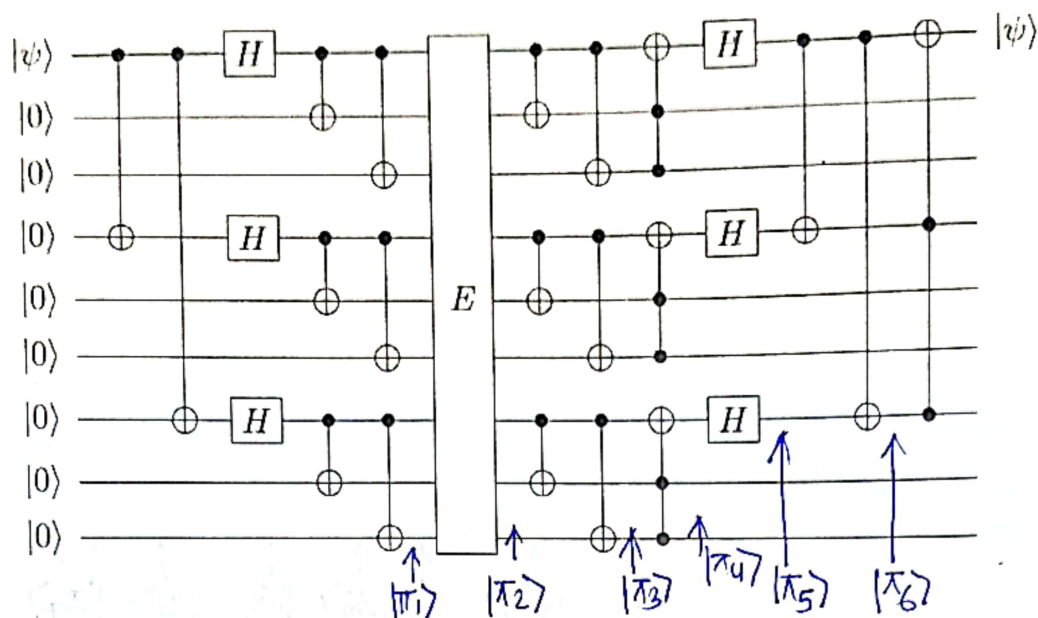
$$|\pi_7\rangle = \frac{1}{\sqrt{2}} \alpha (|107\rangle \otimes |1117\rangle) \otimes (|107\rangle \otimes |1007\rangle) \otimes (|107\rangle \otimes |1007\rangle) \\ + \frac{1}{\sqrt{2}} \beta (|107\rangle \otimes |1117\rangle) \otimes (|107\rangle \otimes |1007\rangle) \otimes (|107\rangle \otimes |1007\rangle)$$

$$|\pi_8\rangle = \frac{1}{\sqrt{2}} \alpha (|107\rangle \otimes |1117\rangle) \otimes (|107\rangle \otimes |1007\rangle) \otimes (|107\rangle \otimes |1007\rangle) \\ + \frac{1}{\sqrt{2}} \beta (|107\rangle \otimes |1117\rangle) \otimes (|107\rangle \otimes |1007\rangle) \otimes (|107\rangle \otimes |1007\rangle)$$

Discarding all other 8 bits, we get $|\psi\rangle = \alpha |107\rangle + \beta |117\rangle$



Single
Phase-flip
errors



$$|\pi_1\rangle = \frac{1}{2\sqrt{2}} \alpha (|1000\rangle + |1111\rangle)^{\otimes 3} + \frac{1}{2\sqrt{2}} \beta (|1000\rangle - |1111\rangle)^{\otimes 3}$$

$$|\pi_1\rangle = \frac{1}{2\sqrt{2}} \alpha (|1000\rangle + |1111\rangle)^{\otimes 3} + \frac{1}{2\sqrt{2}} \beta (|1000\rangle - |1111\rangle)^{\otimes 3}$$

$$|\pi_1\rangle = \frac{1}{2\sqrt{2}} \alpha (|1000\rangle + |1111\rangle)^{\otimes} (|1000\rangle + |1111\rangle)^{\otimes 2} + \frac{1}{2\sqrt{2}} \beta (|1000\rangle - |1111\rangle)^{\otimes} (|1000\rangle - |1111\rangle)^{\otimes 2}$$

$$|\pi_2\rangle = \frac{1}{2\sqrt{2}} \alpha (|1000\rangle - |1111\rangle)^{\otimes} (|1000\rangle + |1111\rangle)^{\otimes 2} + \frac{1}{2\sqrt{2}} \beta (|1000\rangle + |1111\rangle)^{\otimes} (|1000\rangle - |1111\rangle)^{\otimes 2}$$

$$|\pi_3\rangle = \frac{1}{2\sqrt{2}} \alpha (|1000\rangle - |1111\rangle)^{\otimes} (|1000\rangle + |1111\rangle)^{\otimes 2} + \frac{1}{2\sqrt{2}} \beta (|1000\rangle + |1111\rangle)^{\otimes} (|1000\rangle - |1111\rangle)^{\otimes 2}$$

$$|\pi_4\rangle = \frac{1}{2\sqrt{2}} \alpha (|1000\rangle - |1111\rangle)^{\otimes} (|1000\rangle + |1111\rangle)^{\otimes 2} + \frac{1}{2\sqrt{2}} \beta (|1000\rangle + |1111\rangle)^{\otimes} (|1000\rangle - |1111\rangle)^{\otimes 2}$$

$$|\pi_4\rangle = \alpha (|1000\rangle - |1111\rangle)^{\otimes} (|1000\rangle + |1111\rangle)^{\otimes 2} + \beta (|1000\rangle + |1111\rangle)^{\otimes} (|1000\rangle - |1111\rangle)^{\otimes 2}$$

$$|\pi_5\rangle = \alpha (|1100\rangle)^{\otimes} (|1000\rangle)^{\otimes 2} + \beta (|1000\rangle)^{\otimes} (|1100\rangle)^{\otimes 2}$$

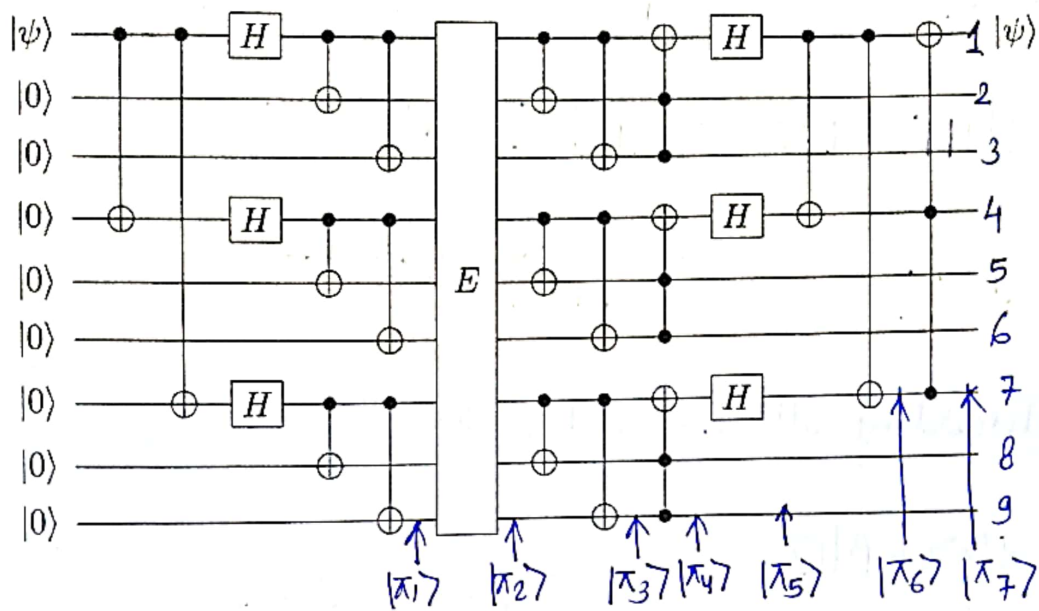
$$|\pi_6\rangle = \alpha (|1100\rangle)^{\otimes} (|1100\rangle)^{\otimes 2} + \beta (|1000\rangle)^{\otimes} (|1100\rangle)^{\otimes 2}$$

$$|\pi_7\rangle = \alpha (|1000\rangle)^{\otimes} (|1100\rangle)^{\otimes 2} + \beta (|1100\rangle)^{\otimes} (|1100\rangle)^{\otimes 2}$$

discarding all other 8 bits we get

$$|\psi\rangle = \alpha |07\rangle + \beta |17\rangle$$

Bit-flip &
phase-flip
both
Y error
 $Y = iXZ$



$$|\pi_1\rangle = \frac{1}{2\sqrt{2}} \alpha (|1000\rangle + |1111\rangle)^{\otimes 3} + \frac{1}{2\sqrt{2}} \beta (|1000\rangle - |1111\rangle)^{\otimes 3}$$

Assuming both phase-flip & bit flip error occurred in the 1st qubit.

After phase-flip in the 1st bit $|\pi_1\rangle$ becomes.

$$= \frac{1}{2\sqrt{2}} \alpha (|1000\rangle - |1111\rangle)^{\otimes 3} + \frac{1}{2\sqrt{2}} \beta (|1000\rangle + |1111\rangle)^{\otimes 3}$$

Then, bit-flip occurs in the first bit so,

$$|\pi_2\rangle = \frac{1}{2\sqrt{2}} \alpha (|1100\rangle - |1011\rangle)^{\otimes 2} + \frac{1}{2\sqrt{2}} \beta (|1100\rangle + |1011\rangle)^{\otimes 2}$$

$$|\pi_3\rangle = \frac{1}{2\sqrt{2}} \alpha (|1111\rangle - |1011\rangle)^{\otimes 2} + \frac{1}{2\sqrt{2}} \beta (|1111\rangle + |1011\rangle)^{\otimes 2}$$

$$|\pi_4\rangle = \frac{1}{2\sqrt{2}} \alpha (|1011\rangle - |1111\rangle)^{\otimes 2} + \frac{1}{2\sqrt{2}} \beta (|1011\rangle + |1111\rangle)^{\otimes 2}$$

$$|\pi_5\rangle = \alpha (|1\rightarrow 1111\rangle)^{\otimes 2} + \beta (|1\rightarrow 1111\rangle)^{\otimes 2}$$

$$|\pi_5\rangle = \alpha |111\rangle (1000)^{\otimes 2} + \beta |011\rangle (1100)^{\otimes 2}$$

$$|\pi_6\rangle = \alpha |111\rangle (1100)^{\otimes 2} + \beta |011\rangle (1000)^{\otimes 2}$$

$$|\pi_7\rangle = \alpha |011\rangle (1100)^{\otimes 2} + \beta |111\rangle (1000)^{\otimes 2}$$

By discarding all other 8 bits we get.

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$$